



DEPARTMENT OF DECISION SCIENCE

FACULTY OF BUSINESS

UNIVERSITY OF MORATUWA

SEMESTER 4

DA2111 - Statistical and Machine Learning

Comprehensive Report

Group Assignment

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Introduction

The Polish Companies Bankruptcy dataset from UCI ML repository([Link](#)) is the selected dataset for group project and its a **Classification** task.

The dataset

Website([Link](#)) provides valuable information/Metadata about dataset. Dataset contains 64 different variables related to financial information of companies. Below figures show information about them.

Additional Variable Information

X1 net profit / total assets
X2 total liabilities / total assets
X3 working capital / total assets
X4 current assets / short-term liabilities
X5 $[(\text{cash} + \text{short-term securities} + \text{receivables} - \text{short-term liabilities}) / (\text{operating expenses} - \text{depreciation})] * 365$
X6 retained earnings / total assets
X7 EBIT / total assets
X8 book value of equity / total liabilities
X9 sales / total assets
X10 equity / total assets
X11 $(\text{gross profit} + \text{extraordinary items} + \text{financial expenses}) / \text{total assets}$
X12 gross profit / short-term liabilities
X13 $(\text{gross profit} + \text{depreciation}) / \text{sales}$
X14 $(\text{gross profit} + \text{interest}) / \text{total assets}$
X15 $(\text{total liabilities} * 365) / (\text{gross profit} + \text{depreciation})$
X16 $(\text{gross profit} + \text{depreciation}) / \text{total liabilities}$
X17 total assets / total liabilities
X18 gross profit / total assets
X19 gross profit / sales
X20 $(\text{inventory} * 365) / \text{sales}$
X21 sales (n) / sales (n-1)
X22 profit on operating activities / total assets
X23 net profit / sales
X24 gross profit (in 3 years) / total assets
X25 $(\text{equity} - \text{share capital}) / \text{total assets}$
X26 $(\text{net profit} + \text{depreciation}) / \text{total liabilities}$
X27 profit on operating activities / financial expenses
X28 working capital / fixed assets
X29 logarithm of total assets
X30 $(\text{total liabilities} - \text{cash}) / \text{sales}$
X31 $(\text{gross profit} + \text{interest}) / \text{sales}$
X32 $(\text{current liabilities} * 365) / \text{cost of products sold}$

X33 operating expenses / short-term liabilities
 X34 operating expenses / total liabilities
 X35 profit on sales / total assets
 X36 total sales / total assets
 X37 (current assets - inventories) / long-term liabilities
 X38 constant capital / total assets
 X39 profit on sales / sales
 X40 (current assets - inventory - receivables) / short-term liabilities
 X41 total liabilities / ((profit on operating activities + depreciation) * (12/365))
 X42 profit on operating activities / sales
 X43 rotation receivables + inventory turnover in days
 X44 (receivables * 365) / sales
 X45 net profit / inventory
 X46 (current assets - inventory) / short-term liabilities
 X47 (inventory * 365) / cost of products sold
 X48 EBITDA (profit on operating activities - depreciation) / total assets
 X49 EBITDA (profit on operating activities - depreciation) / sales
 X50 current assets / total liabilities
 X51 short-term liabilities / total assets
 X52 (short-term liabilities * 365) / cost of products sold
 X53 equity / fixed assets
 X54 constant capital / fixed assets
 X55 working capital
 X56 (sales - cost of products sold) / sales
 X57 (current assets - inventory - short-term liabilities) / (sales - gross profit - depreciation)
 X58 total costs / total sales
 X59 long-term liabilities / equity
 X60 sales / inventory
 X61 sales / receivables
 X62 (short-term liabilities * 365) / sales
 X63 sales / short-term liabilities
 X64 sales / fixed assets

Business Requirement Statement

Objective:

Develop a predictive model to classify companies as either **Bankrupt** or **Non-Bankrupt** based on historical financial data.

Goal:

Proactively identify companies at risk of bankruptcy, enabling stakeholders such as investors, creditors, and management teams to take timely corrective actions to mitigate financial losses and risks.

Key Features:

Leverage relevant financial indicators (e.g., profitability ratios, liquidity ratios, leverage metrics, and operational efficiency metrics) to inform the prediction of a company's bankruptcy status.

Outcome:

Improve the accuracy of bankruptcy predictions, thereby enabling:

- Financial institutions to minimize credit risk.
- Policymakers to design better preventive measures.
- Companies to implement strategies to reduce their bankruptcy risk and ensure sustainable operations.

Understanding the Dataset

The dataset consists of 43405 data points of 65 features including numerical and categorical data. Initial exploration revealed following about the dataset:

```
[ ] # shape of the dataset (rows, columns)
df.shape
(43405, 65)
```

- Categorical Variables: Status (**Target Variable**: 0,1)
- All variables other than target variables are numerical.
- Summary of numerical variables:

	Attr1	Attr2	Attr3	Attr4	Attr5	\
count	43397.000000	43397.000000	43397.000000	43271.000000	4.331600e+04	
mean	0.035160	0.590212	0.114431	6.314702	-3.853466e+02	
std	2.994109	5.842748	5.439429	295.434425	6.124303e+04	
min	-463.890000	-430.870000	-479.960000	-0.403110	-1.190300e+07	
25%	0.003429	0.268980	0.021521	1.049500	-4.908000e+01	
50%	0.049660	0.471900	0.196610	1.569800	-1.034500e+00	
75%	0.129580	0.688320	0.403390	2.787450	5.063425e+01	
max	94.280000	480.960000	28.336000	53433.000000	1.250100e+06	
	Attr6	Attr7	Attr8	Attr9	Attr10	\
count	43397.000000	43397.000000	43311.000000	43396.000000	43397.000000	
mean	-0.056107	0.093478	12.640779	2.652166	0.626868	
std	7.201326	5.713075	505.894281	62.932732	14.670597	
min	-508.410000	-517.480000	-141.410000	-3.496000	-479.910000	
25%	0.000000	0.005776	0.430275	1.018500	0.295470	
50%	0.000000	0.059634	1.070400	1.195350	0.505970	
75%	0.089446	0.150880	2.615700	2.062500	0.709100	
max	543.250000	649.230000	53432.000000	9742.300000	1099.500000	
	Attr55	Attr56	Attr57	Attr58		\
count	...	4.340400e+04	4.327800e+04	43398.000000	4.332100e+04	
mean	...	7.672188e+03	-2.621959e+01	-0.010510	3.002644e+01	
std	...	7.005310e+04	5.327862e+03	13.674072	5.334454e+03	
min	...	-1.805200e+06	-1.108300e+06	-1667.300000	-1.986900e+02	
25%	...	2.755425e+01	9.348500e-03	0.014649	8.753200e-01	
50%	...	1.088350e+03	5.294300e-02	0.119670	9.509600e-01	
75%	...	4.993325e+03	1.290975e-01	0.284605	9.926400e-01	
max	...	6.123700e+06	2.931500e+02	552.640000	1.108300e+06	
	Attr59	Attr60	Attr61	Attr62	Attr63	\
count	43398.000000	4.125300e+04	43303.000000	4.327800e+04	43271.000000	
mean	1.333288	4.480858e+02	17.033202	1.502328e+03	9.343074	
std	122.104445	3.234560e+04	553.049406	1.392667e+05	124.177354	
min	-327.970000	-1.244000e+01	-12.656000	-2.336500e+06	-1.543200	
25%	0.000000	5.545500e+00	4.510150	4.214400e+01	3.097650	
50%	0.006366	9.791700e+00	6.636300	7.132600e+01	5.087600	
75%	0.236052	2.018100e+01	10.394500	1.172200e+02	8.598850	
max	23853.000000	4.818700e+06	108000.000000	2.501600e+07	23454.000000	
	Attr64					
count	42593.000000					
mean	72.788592					
std	2369.339482					
min	-10677.000000					
25%	2.176800					
50%	4.282500					
75%	9.776200					
max	294770.000000					

- Summary of Target variable:

```

Column: class
Unique values: ['0', '1']
Categories (2, object): ['0', '1']
Value counts:
class
0    41314
1     2091
Name: count, dtype: int64

```

- All numerical variables were skewed:

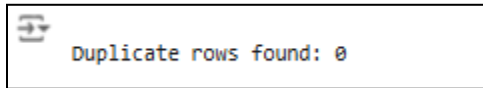
▶	Skewness of Attr13: 117.99900791343107
▶	Skewness of Attr14: 18.901859305941844
▶	Skewness of Attr15: 15.890411673636379
▶	Skewness of Attr16: 57.083841644171315
▶	Skewness of Attr17: 76.88764774260332
▶	Skewness of Attr18: 18.74042517670499
▶	Skewness of Attr19: 158.56048079712332
▶	Skewness of Attr20: 207.90548090904173
▶	Skewness of Attr21: 112.78175518718228
▶	Skewness of Attr22: 58.24399186289683
▶	Skewness of Attr23: 161.770766594348
▶	Skewness of Attr24: 45.77184076746712
▶	Skewness of Attr25: 61.634133567517466
▶	Skewness of Attr26: 58.73515234918379
▶	Skewness of Attr27: 71.8736857581465
▶	Skewness of Attr28: 89.29274881702145
▶	Skewness of Attr29: -0.07895449408765755
▶	Skewness of Attr30: 164.251536280406
▶	Skewness of Attr31: 158.94180652129432
▶	Skewness of Attr32: 151.81164932253276
▶	Skewness of Attr33: 154.24979734804077
▶	Skewness of Attr34: 148.52991877841205
▶	Skewness of Attr35: 60.2368301861186
▶	Skewness of Attr36: 108.84609363196742
▶	Skewness of Attr37: 100.00757869786334
▶	Skewness of Attr38: 48.53409963347438
▶	Skewness of Attr39: -160.4618218336284
▶	Skewness of Attr40: 99.4397639895413
▶	Skewness of Attr41: 206.25748237139996
▶	Skewness of Attr42: 26.59483057343067
▶	Skewness of Attr43: 203.5375178092618
▶	Skewness of Attr44: 200.24578210801573
▶	Skewness of Attr45: 60.415027126247395
▶	Skewness of Attr46: 155.71169781315817
▶	Skewness of Attr47: 158.5622014582119
▶	Skewness of Attr48: 27.119388046075397
▶	Skewness of Attr49: -184.558479071145
▶	Skewness of Attr50: 142.9989740188637
▶	Skewness of Attr51: 82.14764579777824
▶	Skewness of Attr52: 125.34426504467994
▶	Skewness of Attr53: 104.93735482869778
▶	Skewness of Attr54: 104.00437017003172
▶	Skewness of Attr55: 37.91365197830727
▶	Skewness of Attr56: -207.98424458415138
▶	Skewness of Attr57: -71.7288293281093
▶	Skewness of Attr58: 207.0402992787116
▶	Skewness of Attr59: 177.75601639225007
▶	Skewness of Attr60: 122.92470587693268
▶	Skewness of Attr61: 176.2055561516122
▶	Skewness of Attr62: 148.96359737279323
▶	Skewness of Attr63: 163.36689114874628
▶	Skewness of Attr64: 85.11684182602693

All Numerical variables are highly skewed.

Data Preprocessing

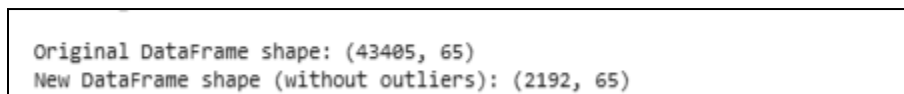
Removing Duplicates

There were no duplicated data rows.



Outlier Handling

Checked outliers using box plots and IQR Method. Dropped outliers from dataset using IQR Method. Removing Outliers caused a loss of substantial number of data rows. After removing outliers, dataset contained 2192 observations.

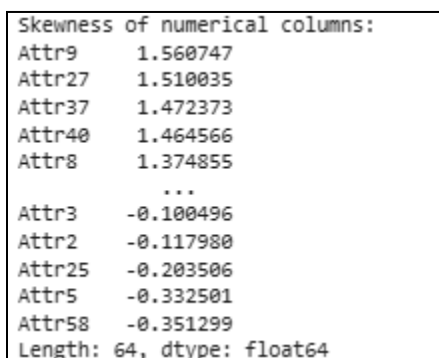


Encoding Categorical Variables

Target Variable which was only categorical variable is already encoded.

Variable Transformations

Log transformed numerical variables to reduce skewness. As below figure shows, skewness of variables has decreased significantly after log transformation and also removing outliers.



Standardizing the dataset

Standardized dataset and below figure shows values in head and tail after standardizing.

	Attr1	Attr2	Attr3	Attr4	Attr5	Attr6	Attr7	\
62	-0.607125	1.554219	-1.676254	-1.196444	-1.022538	-1.638178	-0.405787	
66	0.201559	1.425682	-0.847887	-0.756419	-2.579372	-0.609399	0.307427	
89	0.919577	0.099562	0.082373	0.218068	-0.071644	-1.256518	1.492683	
91	1.504131	-0.155370	-1.606270	-1.382964	-1.480621	0.217054	1.051612	
115	1.961615	0.775124	-0.970189	-0.937268	-0.453925	-0.609399	1.441811	

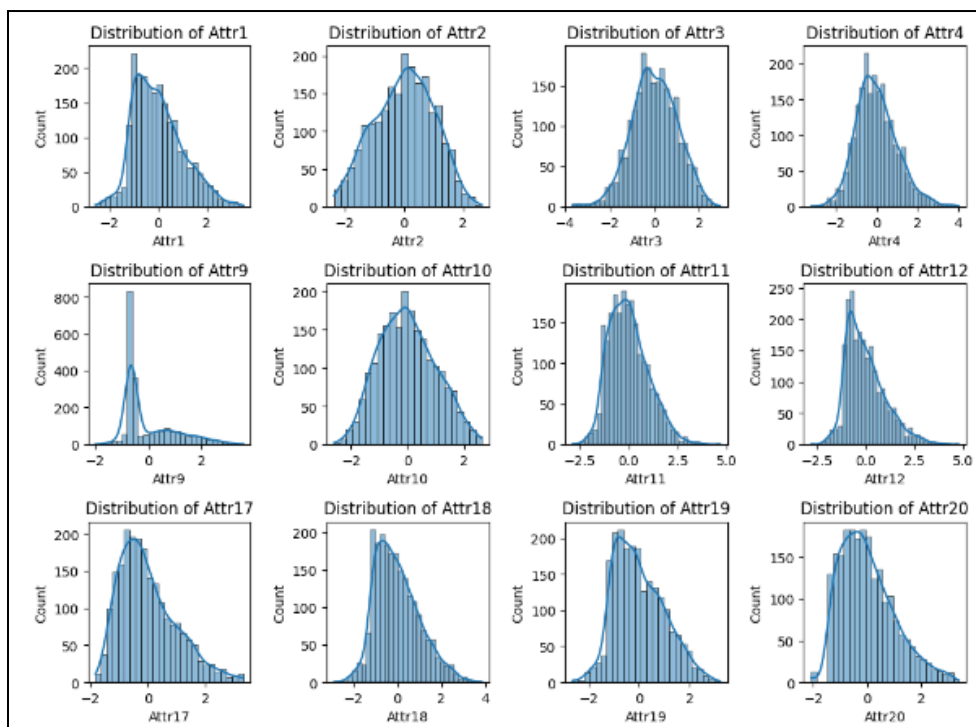
	Attr8	Attr9	Attr10	...	Attr56	Attr57	Attr58	\
62	-1.281886	1.137346	-1.402520	...	0.054967	-0.255862	0.593426	
66	-1.191620	0.560571	-1.271564	...	0.542121	0.871435	-0.209632	
89	-0.108045	0.527456	0.097438	...	2.560573	0.689508	-1.316342	
91	-0.573550	-0.605485	-0.905534	...	0.428508	2.181659	-0.640833	
115	-0.695950	0.502645	-0.599969	...	0.231228	2.310345	-1.192261	

Exploratory Data Analysis(EDA)

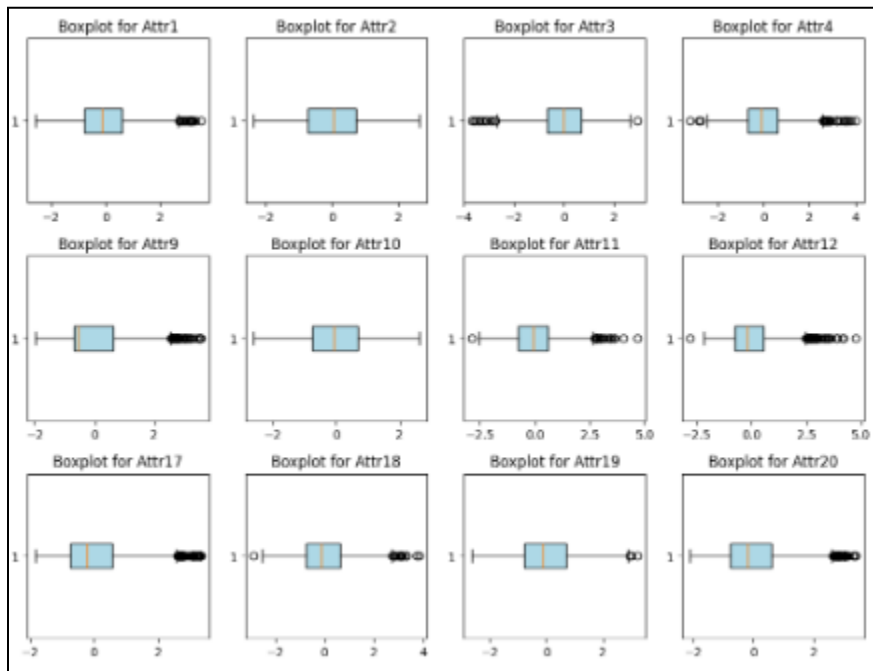
EDA

Various visualization techniques were employed to explore the data and identify relationships:

- Histograms: Displayed the distribution of numerical features.



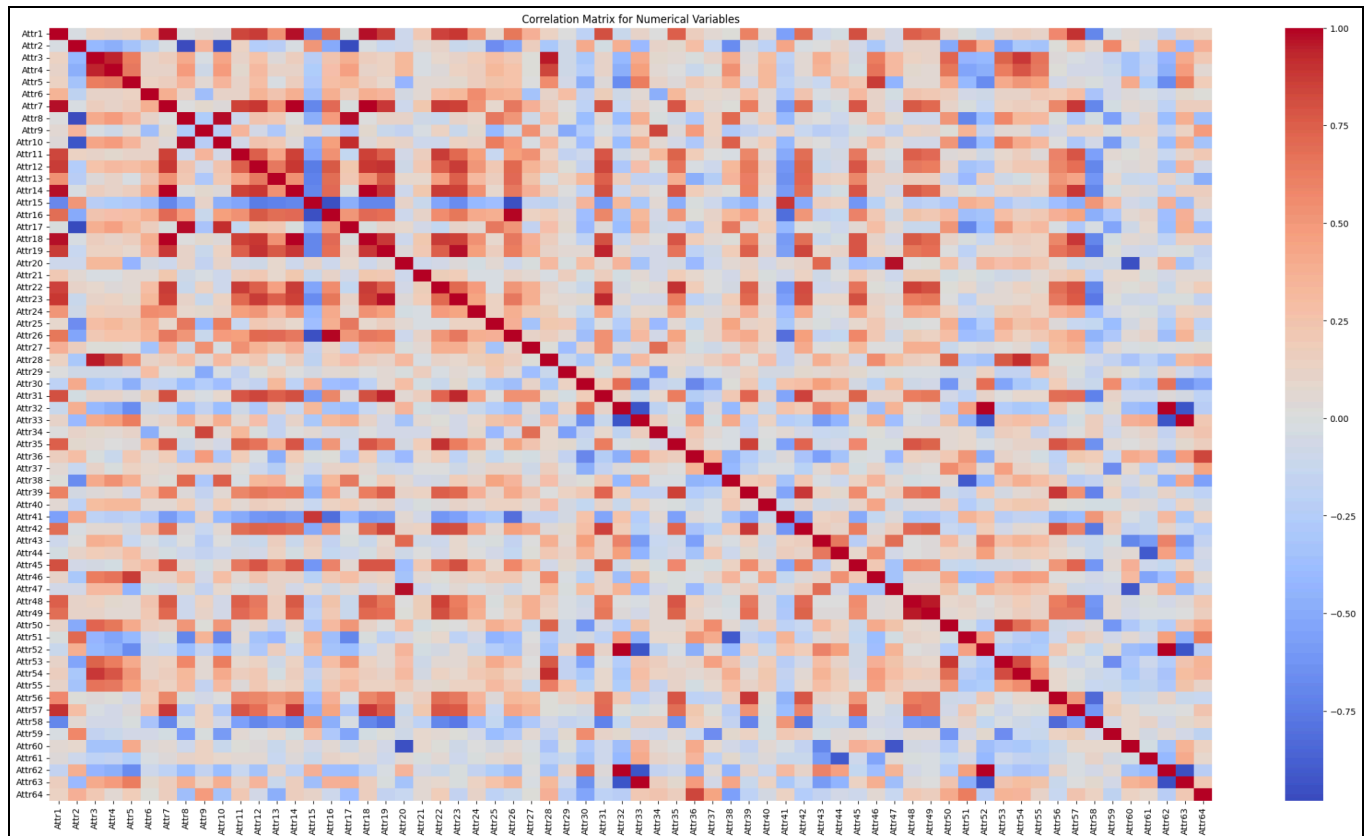
- Box Plots: Shows the distribution of numerical features



- Contingency table for target variable

```
Overall Normalized Contingency Table:
col_0    count
row_0
b'0'     0.930626
b'1'     0.069374
```

- Correlation Matrix: Visualized to detect relationships between features and potential multicollinearity.



Model Selection And Training

Train, validate and test sets splitting

Splitting dataset to 3 sets as training, validate and test set. test set used only once while final evaluation.

```

➡ Training set: (1515, 64) (1515,)
  Validation set: (325, 64) (325,)
  Evaluation set: (325, 64) (325,)

```

Class Balancing

Target variable was unbalanced. Used SMOTE technique to balance classes in **training set**.



```
Class distribution before SMOTE: Counter({'0': 1440, '1': 71})  
Class distribution after SMOTE: Counter({'0': 1440, '1': 1440})
```

Chosen Models:

- **Logistic Regression:** Used Logistic Regression as a baseline model.
- **Decision Tree:** A Decision Tree was chosen because it can model non-linear relationships and capture feature interactions.
- **Random Forest:** Used Random Forest because it reduces overfitting by averaging multiple trees, making it robust for imbalanced datasets, even after applying SMOTE.
- **XGBoost:** XGBoost was selected because it handles complex data patterns.

Evaluation Metrics:

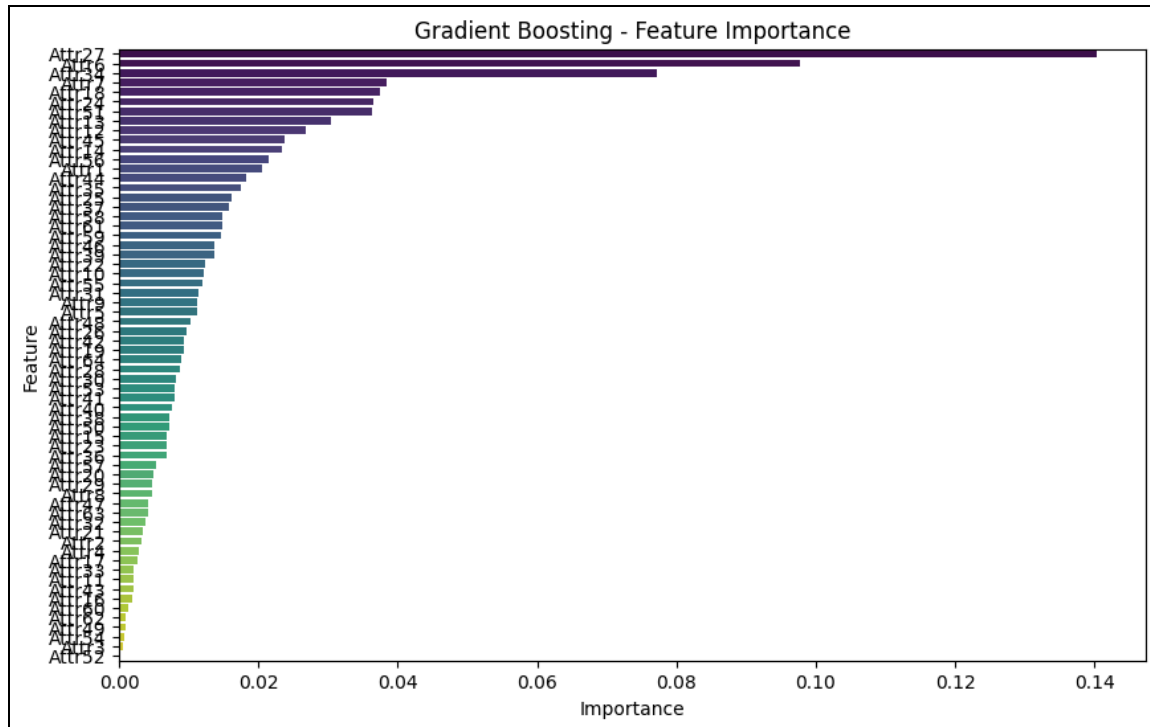
- **Accuracy:** The proportion of correct predictions made by the model.
- **Precision and Recall:** Assessed to understand the trade-off between false positives and false negatives.
- **F1-Score:** A metric that provides a balance between precision and recall.
- **AUC-ROC:** Evaluated to understand the model's ability to distinguish between positive and negative classes.

Performance of features(Tree based model -XGboost)

Key important features for predicting bankruptcy from the best model:

- profit on operating activities / financial expenses
- retained earnings / total assets
- operating expenses / total liabilities
- EBIT / total assets
- gross profit / total assets

Which means company's profitability, financial stability, and efficiency in managing resources will be good indicators for bankruptcy.

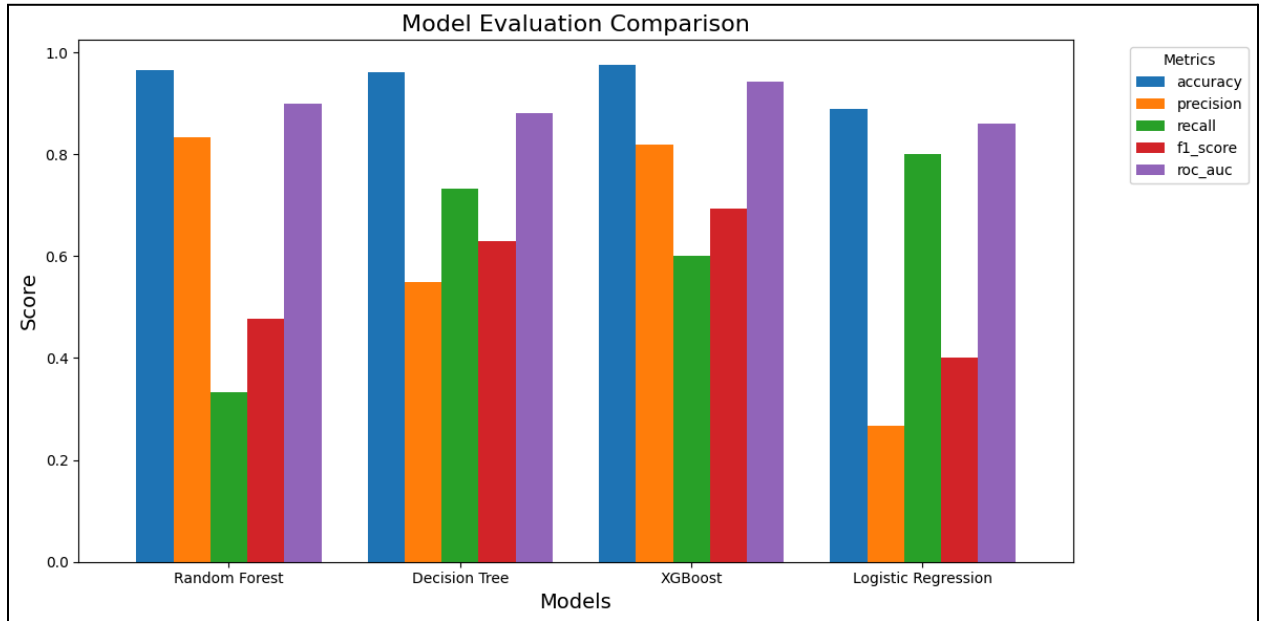


Evaluation of results of models

	Initial					\
	accuracy	precision	recall	f1_score	roc_auc	
Logistic Regression	0.858462	0.196078	0.666667	0.303030	0.827097	
Decision Tree	0.932308	0.360000	0.600000	0.450000	0.774194	
Random Forest	0.975385	0.818182	0.600000	0.692308	0.974731	
XGBoost	0.972308	0.687500	0.733333	0.709677	0.983226	
	Retrained					\
	accuracy	precision	recall	f1_score	roc_auc	
Logistic Regression	0.855385	0.166667	0.533333	0.253968	0.810753	
Decision Tree	0.923077	0.343750	0.733333	0.468085	0.826667	
Random Forest	0.969231	0.727273	0.533333	0.615385	0.976129	
XGBoost	0.975385	0.733333	0.733333	0.733333	0.985376	
	Final Evaluation					
	accuracy	precision	recall	f1_score	roc_auc	
Logistic Regression	0.889231	0.266667	0.800000	0.400000	0.860645	
Decision Tree	0.960000	0.550000	0.733333	0.628571	0.879785	
Random Forest	0.966154	0.833333	0.333333	0.476190	0.899140	
XGBoost	0.975385	0.818182	0.600000	0.692308	0.941720	

Conclusion

1. Best Model:



Based on the final evaluation results, **XGBoost** emerges as the best model with a high overall performance across all metrics. It achieved:

- **Accuracy:** 97.54%
- **Precision:** 81.82%
- **Recall:** 60.00%
- **F1 Score:** 69.23%
- **ROC-AUC:** 94.17%

These results indicate a strong balance between precision and recall, along with superior generalization.

2. Rationale for Selecting XGBoost:

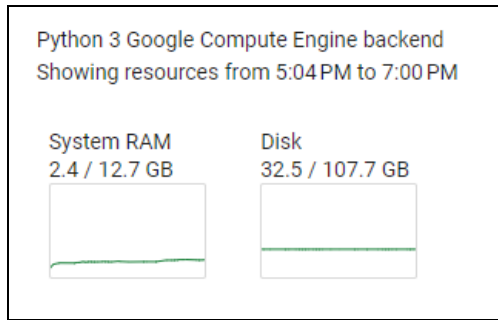
- **High ROC-AUC:** XGBoost consistently outperformed other models in capturing the trade-off between sensitivity and specificity.
- **Balanced Recall and Precision:** This is crucial for the imbalanced dataset since identifying minority class instances is as important as minimizing false positives.

- **Consistency Across Evaluation Phases:** XGBoost maintained strong performance from initial training to the final evaluation, showing robustness and adaptability.
3. **Overfitting Observations:**
- **Logistic Regression:** The drop in performance during retraining and evaluation suggests underfitting or that it cannot handle the complexity of the dataset despite SMOTE balancing.
 - **Decision Tree:** While it improved in recall during retraining, its overall performance is less consistent, suggesting susceptibility to overfitting during training.
 - **Random Forest:** While it achieved good results initially, the decline in recall during evaluation indicates potential overfitting or difficulty generalizing minority class predictions.
 - **XGBoost:** XGBoost showed minimal decline in metrics between retraining and final evaluation, suggesting it is less prone to overfitting.
4. **Other Observations:**
- **Logistic Regression:** Although simple and interpretable, it struggled with capturing complex patterns in the dataset even after SMOTE, as indicated by its lower recall and F1 scores.
 - **Tree-Based Models:** The performance of Decision Tree and Random Forest confirms their ability to handle non-linearity in the data, though their tendency toward overfitting requires careful tuning.
 - **Class Imbalance Handling:** SMOTE proved effective in balancing the dataset, but its impact varied across models. Ensemble-based methods like Random Forest and XGBoost leveraged it better than others.

Infrastructure Details and Analysis

Infrastructure

- Used GoogleColab free version with provided resources to complete project.



- To access to colab used personal laptops of group members,
 - Laptop - Intel i5 CPU, 8GB Ram, Nvidia GeForce GPU
 - Laptop - Intel i7 CPU, 16GB Ram, Intel Iris Xe UHD graphics.